

Relational Schema

Appointment(appointment_id, *client_id*, *staff_id*, *service_id*, appointment_date_time, appointment_type, status, phone_number)
appointment_id → client_id, staff_id, service_id, appointment_date_time, appointment_type, status, phone_number

Client(client_id, name, email, address, date_registered)
client_id → name, email, address, date_registered

Case(case_id, *client_id*, *case_worker_id*, case_description, status, date_opened, date_closed)
case_id → client_id, case_worker_id, case_description, status, date_opened, date_closed

CaseWorker(case_worker_id, name, email, address, phone_number)
case_worker_id → name, email, address, phone_number

Referral(referral_id, *case_id*, *service_id*, referral_date, address, phone_number)
referral_id → case_id, service_id, referral_date, address, phone_number

AdminStaff(staff_id, *organization_id*, name, role, email)
staff_id → organization_id, name, role, email

Service(service_id, *organization_id*, email, address, phone_number)
service_id → organization_id, email, address, phone_number

Shelter(service_id, shelter_address, shelter_capacity, current_occupancy, max_stay_duration)
service_id → shelter_address, shelter_capacity, current_occupancy, max_stay_duration

Organization(organization_id, name, email, address, organization_type)
organization_id → name, email, address, organization_type

Counseling(service_id, counseling_type, session_length, session_format, license_required)
service_id → counseling_type, session_length, session_format, license_required

AppointmentNotes(note_id, *appointment_id*, note_text, created_date)
note_id → appointment_id, note_text, created_date

ServiceSchedule(schedule_id, *service_id*, day_of_week, start_time, end_time)
schedule_id → service_id, day_of_week, start_time, end_time

Each relation's primary key defines all other attribute values. Because the determinant in every functional dependence is a superkey, each relation meets the definition of BCNF.

